Hormones

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Objectives

- ✓ *Definition*
- ✓ *Classification*
- ✓ *Mechanism of action*

ENDOCRINOLOGY

- The nervous system and endocrine system are the major control mechanisms that integrate the functions of the tissues in the body.

- **Definition:**

  " Study of **hormones** secreted by endocrine glands, is called endocrinology"

- **Endocrinology = Intercellular Chemical Communication and information transfer**.

HORMONES

Introduction:

- Most glands of the body deliver their secretions by means of ducts, are called as exocrine glands.
- There are few other glands that produce **chemical substance** that they directly secrete into the bloodstream for transmission to various target tissues.
- These are ductless or endocrine glands.
- ***"The secretions of endocrine glands are called as hormones"*** **.**

Definition

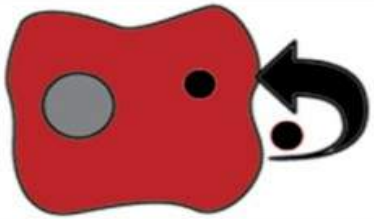
- The word hormone is derived from a Greek word which means “**to excite**” **or “setting in motion”.**

“ It is a chemical substance that is synthesized by a group of cells, transported through blood and is carried to distant target organs and tissues to exerts a physiological control effect on other cells of body to modify their structures and functions”.

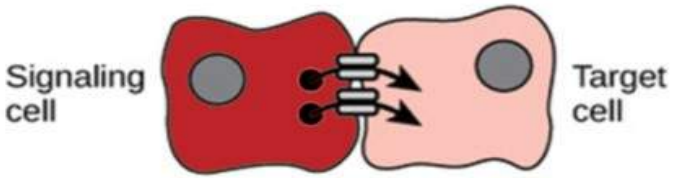
- The hormones control diverse metabolic processes, despite their varying actions and different specificities depending on the target organ.

Forms of Chemical Signaling

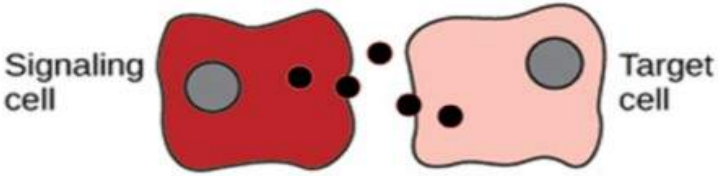
Autocrine	A cell targets itself.



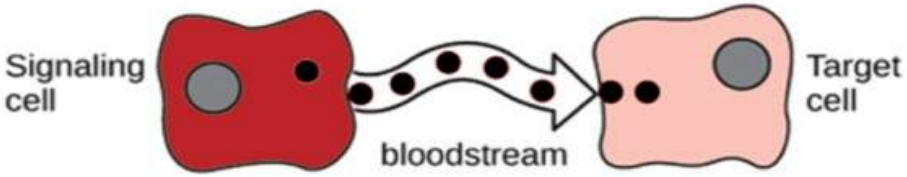
Signaling across gap junctions	A cell targets a cell connected by gap junctions.



Paracrine	A cell targets a nearby cell.



Endocrine	A cell targets a distant cell through the bloodstream.

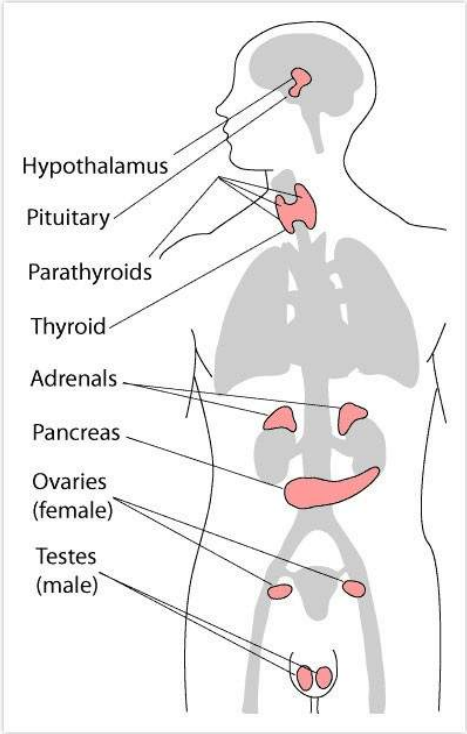


What are the elements of an endocrine system?

- *Sender* = Sending Cell
- *Signal* = Hormone
- *Medium* = Serum
- Receiver= Target cells

The major hormone secreting glands are:

- Pituitary
- Thyroid
- Parathyroid
- Adrenal
- Pancreas
- Ovaries
- Testes



Endocrine Glands and Their Hormones

Origin	Hormones	Function
Hypothalamus	Stimulators Inhibitors (**see Table 2**)	Pituitary hormone regulation
Pituitary, anterior	ACTH FSH Growth hormone Prolactin LH TSH	Adrenal control Gonad regulation Growth regul./milk prod. Milk production Gonad regulation Thyroid control
Pituitary, posterior	ADH Oxytocin	Water conservation Uterus contraction & milk ejection
Thyroid	Thyroxine	Metabolic rate control
Parathyroid	PTH Calcitonin	Calcium regulation Calcium regulation
Gut	Gut hormones	Food digestion
Pancreas	Insulin Glucagon	Nutrient metabolism Nutrient metabolism
Adrenals	Cortisol Aldosterone Epinephrine	Body preservation Salt conservation Stress response
Ovaries	Estradiol Progesterone	Female characteristics
Testes	Testosterone	Male characteristics

Other Endocrine glands

GIT: Few hormones are also produced by certain specialized cells of GI tract and they are called GI Hormones. E.g. **Gastrin, CCK, secretin, somatostatin, ghrelin, and gastrin-releasing peptide (GRP)**

JG cells of Kidney: erythropoietin

Adipose tissue: estrogen from testosterone

Classification of Hormones:

The hormones can be classified **chemically** into four major groups:

- **PEPTIDES:** These are either large proteins or small or medium size peptides. e.g. GH, ACTH, Prolactin, TSH, FSH, LH, Vasopressin, Oxytocin, Insulin, Glucagon and Parathyroid Hormone.
- **STEROIDS:** Aldosterone, cortisol, androgenic hormones, estrogen, progesterone, testosterone and Vitamin D3.
- **AMINES:** Acetylcholine, catecholamines and melatonin.
- **AMINO ACID DERIVATIVES:** These are derived from amino acid tyrosine, e.g. Tetraiodothyronine (T4) (thyroxine) and triiodothyronine.

Classification of Hormones

I. Local: local effects.

II. General: they affect body cells far away from their point of secretions.

Sub-types: two,

- General hormones that affect all cells of body, e.g. growth hormone
- General hormones that affect only target cells because they have specific receptors for hormones e.g. ACTH

REGULATION

The availability of the hormones for a target tissue is regulated by four factors:

- **The rate of synthesis and secretion of the stored hormones.**
- **In some cases, the extent of availability of the specific transport system in the plasma.**
- **Hormone-specific receptors in target cell membranes which differ from tissue to tissue.**
- **The rate of hormone degradation usually by the liver or kidneys.**

REGULATION OF HORMONE SECRETION

Several mechanisms:

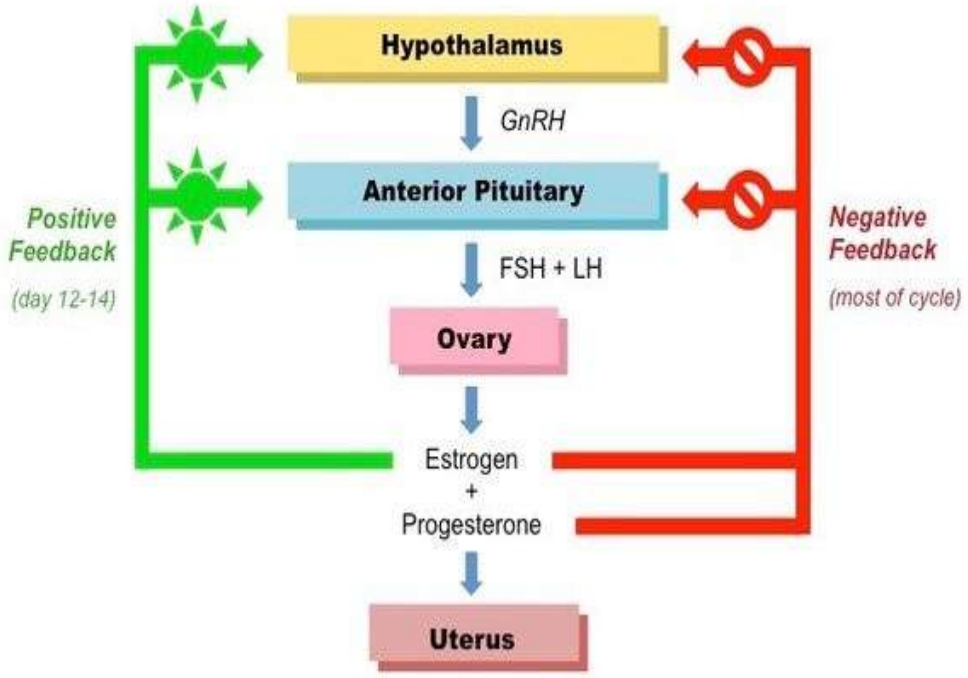
1. Neuroendocrinal control mechanism: Nerve impulses control some endocrine secretions.

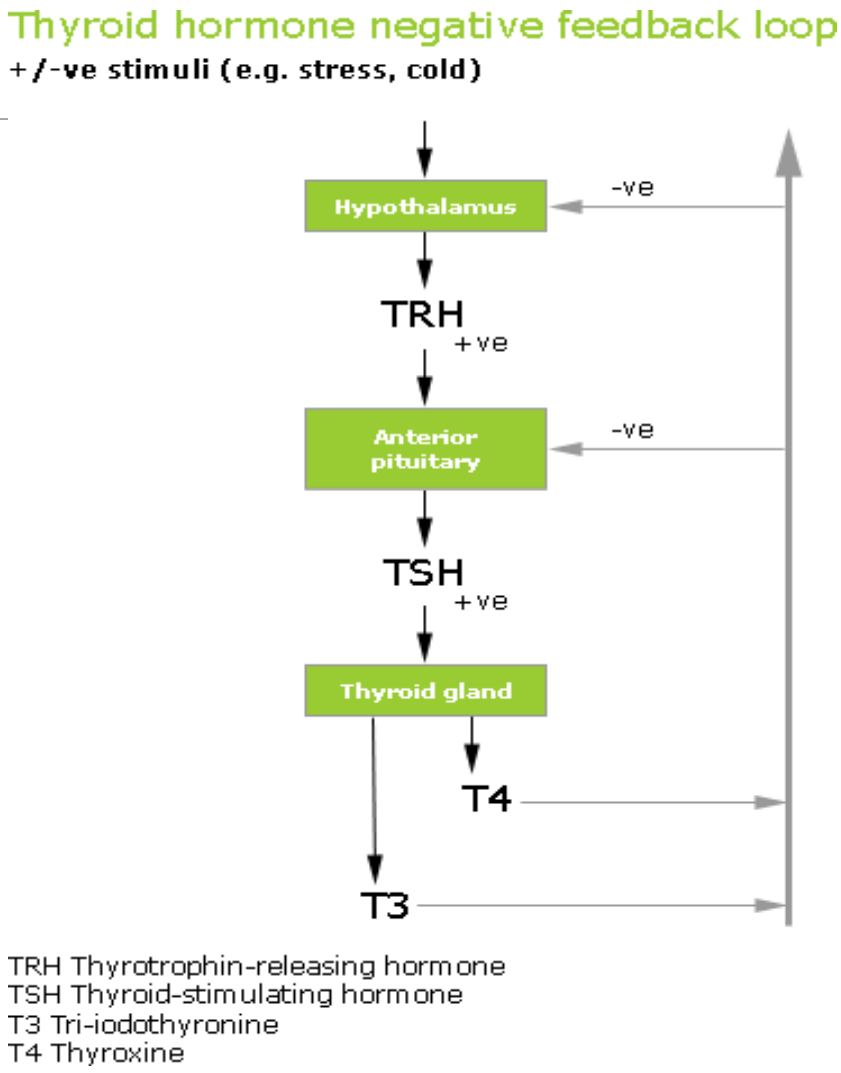
- Examples: Centres in the midbrain, brainstem, etc. can send nerve impulses which react with the hypothalamus to regulate the secretions of pituitary peptide hormones from hypothalamic peptidergic neurons.
- Some of the endocrine releases are controlled by either stimulatory or inhibitory hormones from a controlling gland, e.g. corticosteroids are controlled by corticotropin and thyroid hormones are controlled by thyrotropin from anterior pituitary. The tropins are further regulated by hypothalamic releasing hormones.

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2. Feedback control mechanism:

- Negative feed back mechanism
- Positive feedback system





cont..

3. Endocrine rhythms: There are certain cyclic rhythms associated with the secretion of hormones over a period of time.

- Due to rhythms, the highest and lowest conc. of corticotropin is normally found in the morning and around midnight.
- Growth hormone and prolactin rise in the early hours of deep sleep.
- Cortisol peak is found between 4 AM and 8 AM.
- Endocrine rhythms result from cyclic activities of a biological clock in the limbic system, supplemented by the diurnal light-dark and sleep activity cycles and mediated by the hypothalamus.

MECHANISM OF ACTION OF HORMONES

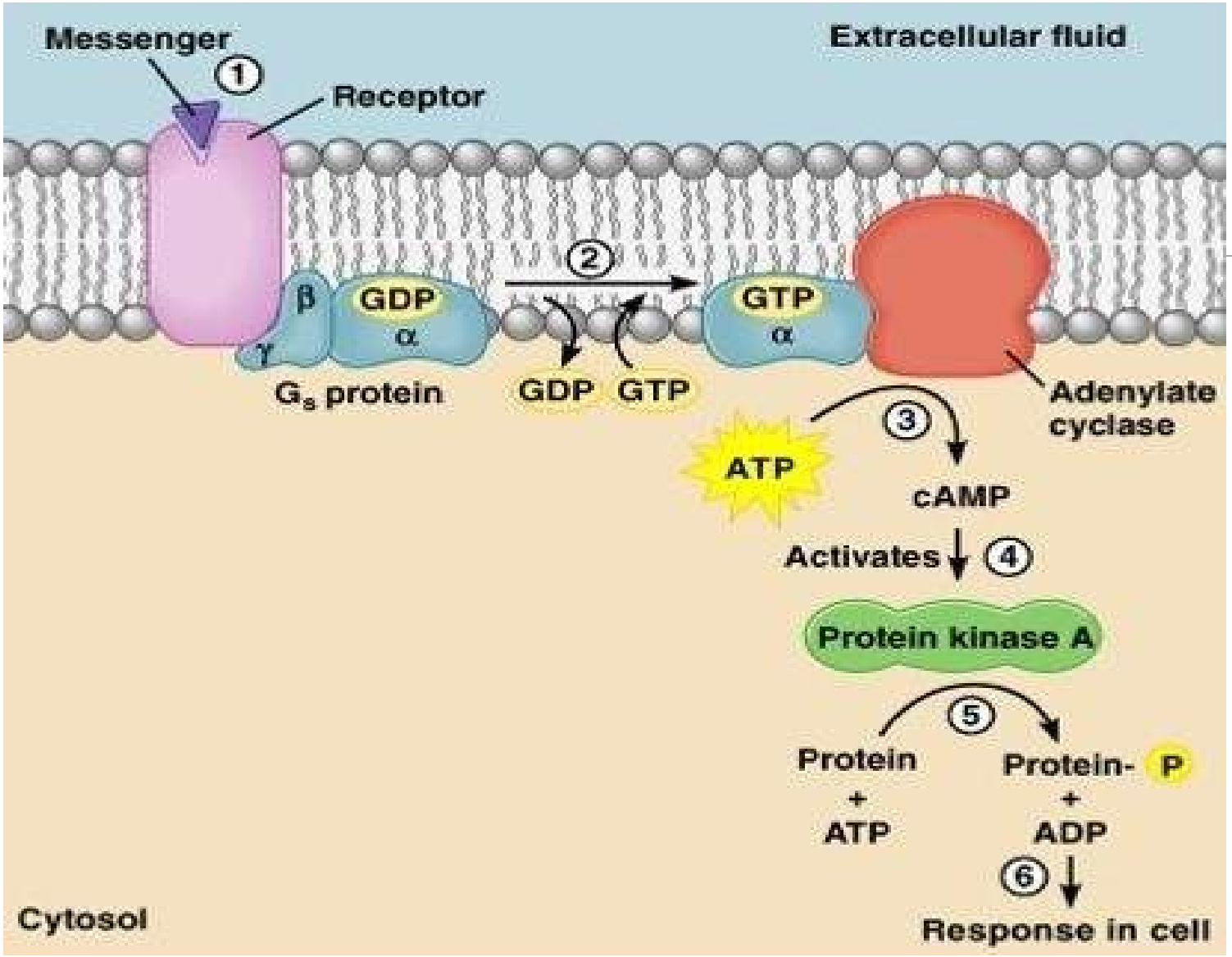
Based on mechanism of action, the hormones may be classified into two:

I. Hormones with cell surface receptors

II. Hormones with intracellular receptors

Hormones with cell surface receptors

- Most protein hormones and catecholamines activate transport of membrane enzyme systems by direct binding to specific receptors on the membrane
  - Hormones bind with cell surface receptors with cAMP as the second messenger. e.g. ACTH, ADH, FSH, Glucagon, Calcitonin and Catecholamines.
  - Hormones having cell surface receptors; cGMP as second messenger. e. g. ANF (atrial natriuretic factor)
  - Hormones having cell surface receptors; second messenger is calcium or phosphatidyl inositol (PIP2). e.g. TRH, GnRH
  - Hormones having cell surface receptors and mediated through tyrosine kinase. e.g. Insulin

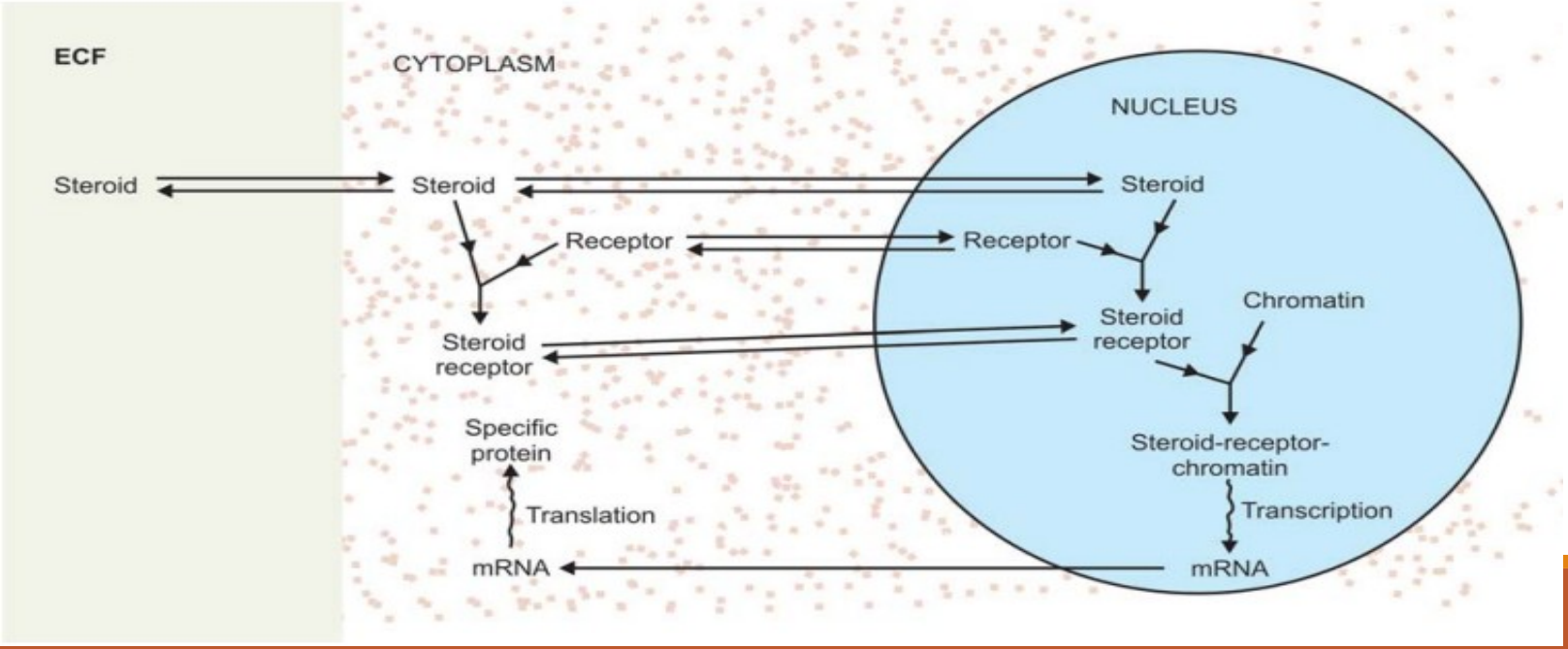

Messenger
1
Receptor
Extracellular fluid
2
β
γ
GDP
α
GTP
α
G_s protein
GDP
GTP
Adenylate cyclase
3
ATP
cAMP
Activates
4
Protein kinase A
5
Protein
+
ATP
Protein-
P
+
ADP
6
Response in cell
Cytosol

Hormones with intracellular receptors

- Hormones that bind to intracellular receptors.
- The hormones in this group include the steroid hormones and thyroid hormones.
  - e.g. Glucocorticoids, Mineralocorticoids, Estrogens, Progesterone, Androgens ,Calcitriol and Thyroxine.
- They diffuse through the plasma membrane and bind to the receptors in the cytoplasm or inside the nucleus.
- The hormone receptor (HR) complex is formed in the cytoplasm. The complex is then translocated to the nucleus.

Mechanism of action of steroid hormones through interaction with nuclear chromatin

- Steroid hormones act mostly by changing the transcription rate of specific genes in the nuclear DNA.
- The receptor-steroid complex is translocated to the nuclear chromatin and binds to a steroid-recognising acceptor site called the **hormone-responsive element (HRE)** of a DNA strand for a specific steroid responsive gene.



Endocrine Functions

- Maintain Internal Homeostasis
- Support Cell Growth
- Coordinate Development
- Coordinate Reproduction
- Facilitate Responses to External Stimuli

Function of a Hormone

Hormones mainly control different metabolic functions of body, e.g.

- ✓ **Rates of chemical reactions:** Hormones are stimulating substances and act as body catalysts.
  - Similarities of Hormone and Enzyme vs Dissimilarities of Hormone and Enzyme
- ✓ **Transport of substances thru cell membrane**
- ✓ **Secretion**
- ✓ **Growth**

General Characteristics of Hormones

- Receptors for hormones:
  - It is the presence of specific receptors for specific hormone explains why hormones act only on certain cells. Sometimes there is a congenital insensitivity of receptors and hormone will not be able to exert its characteristic effect known as Organ insensitivity.
- Half life: varies from a few minutes to several days
- Onset of action
- Types of action
  - Specific
  - Generalized
- Storage: except thyroid hormones few hormones are stored in the endocrine glands
- Release of hormones by passive diffusion, exocytosis.

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- Regulation of secretion

- Pro-hormones and pre-pro-hormones

- Measurement of hormone levels: the circulating levels of hormones can give indication of endocrine gland activity and target organ exposure. Because of the small amounts of the hormones required, blood levels of the hormones are extremely low. In many cases it is ng/µg or mIU, etc

- Production of abnormal hormones and ectopic production of hormones.

- Hormone resistance:
  - Defects in receptors
  - Defects in post receptor effective mechanisms
  - Development of antibodies to hormone
  - Development of antibodies to hormone receptors
  - Absence of target cells.


Thank you!